

Parent Class	Field	Ontology Identifier	Definition	Guidance	Examples	Deprecated Label	Deprecated ID	Version Tracking Label	ID	Description/Guidance
Colour Code Legend										
field name in yellow = required										
field name in purple = recommended										
field name in white = optional										
Database identifiers	specimen collector sample ID	GENEPIO:0001123	The user-defined name for the sample.	Store the collector sample ID. If this number is considered identifiable information, provide an alternative ID. Be sure to store the key that maps between the original and alternative IDs for traceability and follow up if necessary. Every collector sample ID from a single submitter must be unique. If the sample is collected from multiple sites, make it concise, unique and consistent within your lab.	ASDFG123		1.0.0	1.0.0	1.0.0	
	sampling site ID	GENEPIO:0100760	The user-defined identifier assigned to a specific location from which samples are taken.	Store the ID for the site from which a sample was taken. The "site" is user defined (e.g. if may be a building and its environs, a specific entity within an environment). Please use the same site ID for all samples from a given site, regardless of when these samples were taken. Any important changes in site location, should be represented with a new site ID.	Site 12A		1.0.0	1.0.0	1.0.0	
Database identifiers	sampling event ID	GENEPIO:0100761	The user-defined identifier assigned to a specific event during which one or more samples are taken, from one or more sites.	Store the ID for the event during which a sample or samples were taken. For example, an event could be one person taking samples from multiple sites, or multiple people taking samples from one site.	Event 120552.1		1.0.0	1.0.0	1.0.0	
Database identifiers	INSDC Project accession	GENEPIO:0001136	The INSDC (i.e., ENA, NCBI, or DDBJ) accession number of the Project(s) to which the sample belongs.	Store the BioProject accession number. BioProjects are an organizing tool that links together raw sequence data, assemblies, and their associated metadata. Each province will have a BioProject. For example, the BioProject for the National Microbiol Lab, a valid NCBI BioProject accession has prefix PRJNA s.g., PRJNA12345, and is created once at the beginning of a new sequencing project.	PRJNA608651		1.0.0	1.0.0	1.0.0	
	INSDC Sample accession	GENEPIO:0001139	The identifier assigned to a Sample in INSDC (i.e., ENA, NCBI, or DDBJ) archives.	Store the accession returned from the Sample submission. NCBI BioSamples will have the prefix SAMN. ENA have the prefix SAMEA, DDBJ have SAMP.	SAMN14180202, SAMEA00000002, SAMP0000000001		1.0.0	1.0.0	1.0.0	
Sample collection	INSDC sequence read accession	GENEPIO:0101203	The identifier assigned to a sequence in one of the International Nucleotide Sequence Database Collaboration (INSDC) repositories.	Store the accession assigned to the submitted sequence. European Nucleotide Archive (ENA) sequence accessions start with ERR, NCBI-SRA accessions start with SRR, DNA Data Bank of Japan (DDBJ) accessions start with DRR and Genome Sequence Archive (GSA) accessions start with GRR.	ERR123456, DRR123456, GRR123456		1.0.0	1.0.0	1.0.0	
Sample collection	sample collection project name	GENEPIO:0100429	The name of the project/initiative/program for which the sample was collected.	Provide the name of the project and/or the project ID here. If the information is unknown or cannot be provided, leave blank or provide a null value.	Watershed Project (HA-120)		1.0.0	1.0.0	1.0.0	
Sample collection	data steward contact name	GENEPIO:0102221	The name of the individual responsible for the data governance. (meta/data usage and distribution of the data.	Provide the name of the data steward.	Joe Bloggs		1.0.0	1.0.0	1.0.0	
Sample collection	data steward contact email	GENEPIO:0102222	The email address of the individual responsible for the data governance. (meta/data usage and distribution of the data.	The email address can represent a specific individual or lab e.g. johnnyd@lab.ca, or Respi.Lab@lab.ca	WaterTester@acellhy.ca		1.0.0	1.0.0	1.0.0	
Sample collection	organism	GENEPIO:0001191	Taxonomic name of the organism.	Provide the official domain name for the organism(s) e.g. for sample collection, the organism can be a bacterium, a virus, a fungus, a plant, an animal, or a combination of these. See the list of taxonomic names here: ncbi.nlm.nih.gov/taxonomy .	Severe acute respiratory syndrome coronavirus 2		1.0.0	1.0.0	1.0.0	
Sample collection	purpose of sampling	GENEPIO:0001198	The reason that the sample was collected.	The reason a sample was collected may provide information about potential biases in sampling strategy. Provide the purpose of sampling from the picklist in the template. Most likely, the sample was collected for Public health surveillance. The reason why a sample was originally collected may differ from the reason why it was selected for sequencing, which should be indicated in the "purpose of sequencing" field.	Baseline surveillance (random sampling) [GENEPIO:0100005]		1.0.0	1.0.0	1.0.0	
Sample collection	sample collected by organisation name	GENEPIO:0001153	The name of the organization with which the sample collector is affiliated.	The name of the agency should be written out in full, (with minor exceptions) and be consistent across multiple submissions.	Public Health Agency of Canada		1.0.0	1.0.0	1.0.0	
Sample collection	sample collected by contact email	GENEPIO:0001156	The email address of the contact responsible for follow-up regarding the sample.	The email address can represent a specific individual or lab e.g. johnnyd@lab.ca, or Respi.Lab@lab.ca	WaterTester@acellhy.ca		1.0.0	1.0.0	1.0.0	
Sample collection	sample collection data steward name	GENEPIO:0100762	The name of the individual responsible for the data governance. (meta/data usage and distribution of the sample.	Provide the name of the sample collection data steward.	Joe Bloggs		1.0.0	1.0.0	1.0.0	

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	Colour Code Legend field name in yellow = required field name in purple = recommended field name in white = optional									IMPORTANT: Only labels and/or IDs will be deprecated, always with replacement version provided. If a term changes in its meaning, a new term will be created.
				Sample collection date is critical for surveillance and many types of analyses. For samples taken at a single point in time, use this field to capture the date the sample was collected. For samples taken over a continuous period, use this field to capture the sample collection start date. Required granularity includes year, month and day. The date should be provided in ISO 8601 standard format "YYYY-MM-DD".						
Sample collection	sample collection start date	GENEPIO:0001174	The date on which the sample was collected, or sampling began for a continuous sample.				2020-03-16	1.0.0	1.0.0	1.0.0
Sample collection	sample collection end date	GENEPIO:0101071	The date on which sample collection ended for a continuous sample.				2020-03-18	1.0.0	1.0.0	1.0.0
Sample collection	sample collection time duration value	GENEPIO:0100766	The amount of time over which the sample was collected.				4	1.0.0	1.0.0	1.0.0
Sample collection	sample collection time duration unit	GENEPIO:0100767	The units of the time duration measurement of sample collection.		Hour			1.0.0	1.0.0	1.0.0
Sample collection	geo loc name (country)	GENEPIO:0001181	The country of origin of the sample.		Canada			1.0.0	1.0.0	1.0.0
Sample collection	geo loc name (state/province/territory)	GENEPIO:0001185	The state/province/territory of origin of the sample.		Ontario			1.0.0	1.0.0	1.0.0
Sample collection	geo loc name (site)	GENEPIO:0100436	The name of a specific geographical location e.g. Credit River (rather than river).		Credit River			1.0.0	1.0.0	1.0.0
Sample collection	geo loc name (city)	GENEPIO:0001189	The city of origin of the sample.		Vancouver			1.0.0	1.0.0	1.0.0
Sample collection	geo loc latitude	GENEPIO:0100309	The latitude coordinates of the geographical location of sample collection.		38.98 N			1.0.0	1.0.0	1.0.0
Sample collection	geo loc longitude	GENEPIO:0100310	The longitude coordinates of the geographical location of sample collection.		77.11 W			1.0.0	1.0.0	1.0.0
Sample collection	environmental site	GENEPIO:0001232	An environmental location may describe a site in the natural or built environment e.g. hospital, wet market, lake. An environmental location in the natural or built environment, that is proximal to a sampling location and which can impact a sample.		Influent pump station [ENVO:03501465]			1.0.0	1.0.0	1.0.0
Sample collection	proximal environmental site	GENEPIO:0101205	The instrument or container used to collect the sample e.g. grab sampler.		Aquaculture farm [ENVO:03600074]			1.0.0	1.0.0	1.0.0
Sample collection	collection device	GENEPIO:0001234	The process used to collect the sample.		Grab sample [GENEPIO:0100941]			1.0.0	1.0.0	1.0.0
Sample collection	collection method	GENEPIO:0001241	An anatomical material is a substance taken from the body. If applicable, select the standardized term and ontology ID for the anatomical material from the pidlist provided. Multiple values can be provided, separated by a semi-colon.		Grab sampling [GENEPIO:0100953]			1.0.0	1.0.0	1.0.0
Sample collection	anatomical material	GENEPIO:0001211	A substance obtained from an anatomical part of an organism e.g. tissue, blood.		Tissue [UBERON:0000479]; Blood [UBERON:0000178]			1.0.0	1.0.0	1.0.0
Sample collection	anatomical part	GENEPIO:0001214	An anatomical part of an organism e.g. oropharynx.		Fish vent [GENEPIO:0100902]			1.0.0	1.0.0	1.0.0
Sample collection	body product	GENEPIO:0001216	A substance excreted/secreted from an organism e.g. feces, urine, sweat.		Feces [UBERON:0001988]; Urine [UBERON:0001088]			1.0.0	1.0.0	1.0.0

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Presampling activity information	experimental specimen role type	GENEPIO:0100921	The type of role that the sample represents in the experiment.	Samples can play different types of roles in experiments. A sample under study in one experiment may act as a control or be a replicate of another sample in another experiment. This role is determined by the activity from which the sample is collected, and the type of controls, replicates, etc. If this sample acted as an experimental control or a replicate, select a role type from the picklist. If the sample was not a control, leave blank or select "Not Applicable".	Positive experimental control [GENEPIO:0101018]			1.0.0	1.0.0	
Presampling activity information	presampling activity ID	GENEPIO:0102171	The user-defined identifier assigned to a specific event that occurred prior to a sample collection.	Store the ID for the relevant event prior to sample collection, which impacts the sample or sample collection process. For example, a flood event occurring in 2025 is the reason for these samples being collected.	Presampling event 12345			1.0.0	1.0.0	1.0.0
Presampling activity information	presampling activity date	GENEPIO:0102172	The date of the presampling event.	Provide the date of the presampling event. If this date was over a range provide the start date.	2025-01-01			1.0.0	1.0.0	1.0.0
Presampling activity information	presampling activity	GENEPIO:0100433	The activities or variables prior to sample collection that may affect the sample.	If there was an activity that would affect the sample prior to collection (this is different than sample processing), provide the activities by selecting one or more values from the template pick list. If the information is unknown or cannot be provided, leave blank or provide a null value.						
Sample processing	presampling activity details	GENEPIO:0100434	The details of the activities or variables that affected the sample collected.	Briefly describe the presampling activities using free text.	Agricultural activity [ENVO:01001442]			1.0.0	1.0.0	1.0.0
Sample processing	specimen processing	GENEPIO:0001253	Any processing applied to the sample during or after receiving the sample.	Select processes from the picklist that were applied to this sample.	Agricultural waste from large farm contributes waste to the site sampled.			1.0.0	1.0.0	1.0.0
Sample processing	specimen processing details	GENEPIO:0100311	The details of the processing applied to the sample during or after receiving the sample.	Briefly describe the processes applied to the sample.	Filtration [DBI:0302885]			1.0.0	1.0.0	1.0.0
Sample processing	experimental protocol name	GENEPIO:0101029	The name of the overarching experimental methodology that was used to process the biomaterial.	Briefly describe the processes applied to the sample.	5 samples were pooled and further prepared as a single sample during library prep.			1.0.0	1.0.0	1.0.0
Sample processing	experimental protocol url	GENEPIO:0102071	A URL linking to detailed information or documentation about the experimental protocol. This URL should direct users to a source that provides the necessary details to reproduce the experiment that generated the dataset.	Provide the name of the methodology used in your study.	Legionella Testing Protocol 2025			1.0.0	1.0.0	1.0.0
Sample processing	experimental control details	GENEPIO:0100922	The details regarding the experimental control contained in the sample.	Provide the url of the experimental protocol.	https://github.com/vsevolod			1.0.0	1.0.0	1.0.0
Sample processing	nucleic acid extraction method	GENEPIO:0100939	The process used to extract genomic material from a sample.	Provide details regarding the nature of the reference strain used as a control, or what is was used to monitor.	Human coronavirus 229E (HCoV-229E) spiked in sample as process control			1.0.0	1.0.0	1.0.0
Sample processing	nucleic acid extraction kit	GENEPIO:0100772	The kit used to extract genomic material from a sample	Briefly describe the extraction method used.	Direct wastewater RNA capture and purification via the "Savanna Salt, Silica and SARS-CoV-2 (4S)" method v4 found at https://www.protocols.io/view/v4-direct-wastewater-rna-capture-and-purification-36wg5d81ygl5iv4			1.0.0	1.0.0	1.0.0
Sample collection	endogenous control details	GENEPIO:0100923	The description of the endogenous controls included when extracting a sample.	Provide the name of the genomic extraction kit used.	QIAamp PowerFecal Pro DNA Kit			1.0.0	1.0.0	1.0.0
Sample collection	available data types	GENEPIO:0100690	The type of data that is available, that may or may not require permission to access.	Provide the names of endogenous controls that were used as a reference during extraction. If relevant, include others of these controls, as well as whether any controls were expected but not identified in the sample.	Pepper Mkt Mottle Virus (PMMV)			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	available data type details	GENEPIO:0101023	Detailed information regarding other available data types.	This field provides information about additional data types that are available that may provide context for interpretation of the sequence data. Provide a term from the picklist for additional data types that are available. Additional data that are not in the picklist may be specified in the context for this field.	Total coliform count [GENEPIO:0100729]			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	populated area type	GENEPIO:0100778	A type of area that is populated by humans to different degrees.	Use this field to provide free text details describing other available data that may provide context for interpreting genomic sequence data.	Pooled metagenomes containing several different human bacterial taxa (ESBL) bacteria			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	water catchment area human population measurement value	GENEPIO:0100773	The numerical value of the human population measurement from the water catchment area.	Provide the populated area type from the pick list.	Urban [GSSO:011080]			1.0.0	1.0.0	1.0.0
				Where known, provide the numerical value of population size, i.e. the number of people.	10,500			1.0.0	1.0.0	1.0.0

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Environmental conditions and measurements	Colour Code Legend	IMPORTANT: Only labels and/or IDs will be deprecated, always with replacement version provided. If a term changes in its meaning, a new term will be created.								
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Environmental conditions and measurements	water catchment area human population range	GENEPIO:0100774	The human population range of the water catchment that contributes effluent to a wastewater site.	Where catchment population is not well known, provide an estimation of population size by selecting a value from the picklist.	1,000 - 10,000 people			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	presampling weather conditions	GENEPIO:0100780	Weather conditions prior to collection that may affect the sample.	Provide the weather conditions prior to sample collection by selecting a value from the picklist.	Drizzle [ENVO:03501420]			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	sampling weather conditions	GENEPIO:0100779	The state of the atmosphere at a place and time as regards heat, dryness, sunshine, wind, rain, etc.	Provide the weather conditions at the time of sample collection by selecting a value from the picklist.	Rain [ENVO:01001564]			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	precipitation measurement value	GENEPIO:0100911	The amount of water which has fallen during a precipitation process.	Provide the quantity of precipitation in the area leading up to the time of sample collection.	12			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	precipitation measurement unit	GENEPIO:0100912	The units of measurement for the amount of water which has fallen during a precipitation process.	Provide the units of precipitation by selecting a value from the pick list.	inch			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	water depth value	GENEPIO:0100440	The depth of some water from which the sample was collected.	Provide the numerical depth only of water only (without units).	5			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	water depth units	GENEPIO:0101025	The units of measurement for water depth.	Provide the units of measurement for which the depth was recorded by selecting a value from the picklist.	meter (m) [UO:0000008]			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	sediment depth value	GENEPIO:0100697	The depth of some sediment from which the sample was collected.	Provide the numerical depth only of the sediment (without units).	2			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	sediment depth units	GENEPIO:0101026	The units of measurement for sediment depth.	Provide the units of measurement for which the depth was recorded by selecting a value from the picklist.	meter (m) [UO:0000008]			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	pH measurement value	GENEPIO:0001736	The measured pH value indicating the acidity or basicity(alkalinity) of an aqueous solution.	Provide the numerical value of the measured pH.	7.4			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	total daily flow rate measurement value	GENEPIO:0100905	The numerical value of a measured fluid flow rate over the course of a day.	Provide the numerical value of the measured flow rate.	10			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	total daily flow rate measurement unit	GENEPIO:0100906	The units of a measured fluid flow rate over the course of a day.	Provide the units of the measured flow rate by selecting a value from the pick list.	million gallons per day (MGD)			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	instantaneous flow rate measurement value	GENEPIO:0100908	The numerical value of a measured instantaneous fluid flow rate.	Provide the numerical value of the measured flow rate.	25			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	instantaneous flow rate measurement unit	GENEPIO:0100909	The units of a measured instantaneous fluid flow rate.	Provide the units of the measured flow rate by selecting a value from the pick list.	cubic meter per hour (m³/h)			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	turbidity measurement value	GENEPIO:0100783	The numerical value of a measurement of turbidity.	Provide the numerical value of the measured turbidity.	0.02			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	turbidity measurement unit	GENEPIO:0100914	The units of a measurement of turbidity.	Provide the units of the measured turbidity by selecting a value from the pick list.	nephelometric turbidity unit (NTU)			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	conductivity measurement value	GENEPIO:0000034	The numerical value of a measurement of conductivity.	Provide the numerical value of the measured conductivity.	1412			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	conductivity measurement unit	GENEPIO:0100803	The units of a measurement of conductivity.	Provide the units of the measured conductivity by selecting a value from the pick list.	millisiemens per centimeter [GENEPIO:0001759]			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	salinity measurement value	GENEPIO:0100805	The numerical value of a measurement of salinity.	Provide the numerical value of the measured salinity.	35			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	salinity measurement unit	GENEPIO:0100806	The units of a measurement of salinity.	Provide the units of the measured salinity by selecting a value from the pick list.	practical salinity unit (PSU)			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	dissolved oxygen measurement value	GENEPIO:0000035	The numerical value of a measurement of dissolved oxygen.	Provide the numerical value of the measured dissolved oxygen.	5			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	dissolved oxygen measurement unit	GENEPIO:0100784	The units of a measurement of dissolved oxygen.	Provide the units of the measured dissolved oxygen by selecting a value from the pick list.	part per million (ppm)			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	carbonaceous biochemical oxygen demand (CBOD) measurement value	GENEPIO:0100791	The numerical value of a measurement of carbonaceous biochemical oxygen demand (CBOD).	Provide the numerical value of the measured CBOD.	20			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	carbonaceous biochemical oxygen demand (CBOD) measurement unit	GENEPIO:0100792	The units of a measurement of carbonaceous biochemical oxygen demand (CBOD).	Provide the units of the measured CBOD by selecting a value from the pick list.	milligram per liter (mg/L)			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	total suspended solids (TSS) measurement value	GENEPIO:0100794	The numerical value of total suspended solids of a sample.	Provide the numerical value of the measured TSS.	8			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	total suspended solids (TSS) measurement unit	GENEPIO:0100795	The units associated with a value from a total suspended solids (TSS) test.	Provide the units of the measured TSS by selecting a value from the pick list.	percent (%)			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	fecal contamination indicator	GENEPIO:0100814	A gene, virus, bacteria, or substance used to measure the quality of water in regards to fecal contamination.	If a fecal contamination indicator was measured, select it from the picklist.	cAsphage			1.0.0	1.0.0	1.0.0
Environmental conditions and measurements	fecal contamination value	GENEPIO:0100815	The numerical value of a measurement of fecal contamination.	Provide the numerical value of the measured fecal contamination.	10			1.0.0	1.0.0	1.0.0

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Host Information	Colour Code Legend field name in yellow = required								
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	fecal contamination unit	GENEPIO:0100816	The units of a measurement of fecal contamination.	Provide the units of the measured fecal contamination by selecting a value from the picklist.	PCR quantification cycle (Ct value) [UO:0010077]		1.0.0	1.0.0	1.0.0
	host (common name)	GENEPIO:0001386	The commonly used name of the host.	If the sample is directly from a host, either a common or scientific name must be provided (although both can be included) by selecting a value from the picklist. If known, provide the common name.	Chinook salmon [NCBI:Taxon/74940]		1.0.0	1.0.0	1.0.0
Host Information	host (scientific name)	GENEPIO:0001387	The taxonomic, or scientific name of the host.	If the sample is directly from a host, either a common or scientific name must be provided (although both can be included), by selecting a value from the picklist. If known, select the scientific name from the picklist provided.	Homo sapiens [NCBI:Taxon/9606]		1.0.0	1.0.0	1.0.0
Host Information	host age	GENEPIO:0001392	Age of host at the time of sampling.	If known, provide the age of the host. Age-binning is also acceptable.	79		1.0.0	1.0.0	1.0.0
Host Information	host age unit	GENEPIO:0001393	The units used to measure the host's age.	If known, provide the age units used to measure the host's age from the pick list.	year [UO:0000036]		1.0.0	1.0.0	1.0.0
Strain and Isolation Information	host age bin	GENEPIO:0001394	Age of host at the time of sampling, expressed as an age group.	Select the corresponding host age bin from the pick list provided in the template. If not available, provide a null value or leave blank. NOTE: Age bins for animals may differ from those used for humans.	First winter [GENEPIO:0100684]		1.0.0	1.0.0	1.0.0
Strain and Isolation Information	isolate ID	GENEPIO:0001644	The user-defined identifier for the isolate, as provided by the laboratory that originally isolated the isolate.	Provide the isolate ID created by the lab that first isolated the isolate (i.e. the original isolate ID). If the information is unknown or cannot be provided, leave blank or provide a null value. If only an alternate isolate ID is known (e.g. the ID from your lab, if your lab did not isolate the isolate from the original sample), make assure to include it in the alternative_isolate_ID field.	SA20131043		1.0.0	1.0.0	1.0.0
Strain and Isolation Information	microbiological method	GENEPIO:0100454	The laboratory method used to grow, prepare, and/or isolate the microbial isolate.	Provide the name and version number of the microbiological method. The ID of the method is also acceptable if the ID can be linked to the laboratory that created the procedure.	MF+PB-30		1.0.0	1.0.0	1.0.0
Strain and Isolation Information	strain ID	GENEPIO:0100455	The strain identifier.	If the isolate represents or is derived from, a lab reference strain or strain from a type culture collection, provide the strain identifier.	K12		1.0.0	1.0.0	1.0.0
Sequence Information	isolated by organisation name	GENEPIO:0100461	The name of the agency, organization or institution with which the individual who performed the isolation procedure is affiliated.	Provide the name of the agency, organization or institution that isolated the original isolate in full (avoid abbreviations). If the information is unknown or cannot be provided, leave blank or provide a null value.	Public Health Agency of Canada (PHAC) [GENEPIO:0100551]		1.0.0	1.0.0	1.0.0
Sequence Information	purpose of sequencing	GENEPIO:0001445	The reason that the sample was sequenced.	The reason why a sample was originally collected may differ from the reason why it was selected for sequencing. The reason a sample was sequenced may provide information about potential biases in sequencing strategy. Provide the purpose of sequencing from the picklist in the template. The reason for sample collection should be indicated in the "purpose of sampling" field.	Domestic travel surveillance [GENEPIO:0100013]		1.0.0	1.0.0	1.0.0
Sequence Information	assay type	OBH0000070	The experimental process that was used in to quantify, amplify or identify the sequence of a biomaterial.	Provide the name of the sequencing technology used in your study, or alternatively PCR assay, by selecting a value from the picklist. If unsure refer to the protocol documentation, or provide a null value.	Amplikon sequencing assay [OBH:0002767]		1.0.0	1.0.0	1.0.0
Sequence Information	sequencing date	GENEPIO:0001447	The date the sample was sequenced.	ISO 8601 standard "YYYY-MM-DD".	2020-06-22		1.0.0	1.0.0	1.0.0
Sequence Information	sequenced by organisation name	GENEPIO:0100416	The name of the agency, organization or institution responsible for sequencing the isolate's genome.	Provide the name of the agency, organization or institution that performed the sequencing in full (avoid abbreviations). If the information is unknown or cannot be provided, leave blank or provide a null value.	Public Health Agency of Canada (PHAC) [GENEPIO:0100551]		1.0.0	1.0.0	1.0.0
Sequence Information	sequenced by contact email	GENEPIO:0100471	The email address of the contact responsible for follow-up regarding the sequence.	Provide the email associated with the listed contact. As personnel turnover may render an individual's email obsolete, it is more preferable to provide an address and position of lab, to ensure accuracy of information and institutional memory. If the information is unknown or cannot be provided, leave blank or provide a null value.	abcsmith@lab.ca		1.0.0	1.0.0	1.0.0
Sequence Information	sequence submitted by	GENEPIO:0001159	The name of the agency that submitted the sequence to a database.	The name of the agency should be written out in full, (with minor exceptions) and be consistent across multiple submissions. For Canadian institutions submitting specimens rather than sequencing data, please put the "National Microbiology Laboratory (NML)".	Public Health Ontario (PHO)		1.0.0	1.0.0	1.0.0

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	Colour Code Legend									
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	field name in yellow = required									
	field name in white = optional									
Sequence information	sequencing protocol	GENEPIO:0001454	The protocol or method used for sequencing.	Provide the name and version of the procedure or protocol used for sequencing. You can also provide a link to a protocol online.	https://www.protocols.io/view/mov-2019s-sequencing-protocol-b6muk6w2?version=1&amplifier=1			1.0.0	1.0.0	
Sequence information	sequencing instrument	GENEPIO:0001452	The model of the sequencing instrument used.	Provide the model sequencing instrument by selecting a value from the template pick list. If the information is unknown or cannot be provided, leave blank or provide a null value.				1.0.0	1.0.0	1.0.0
Sequence information	library ID	Sequence information	The user-specified identifier for the library used for sequencing.	The library name should be unique, and can be an alphanumeric string. It should not contain a hyphen (-) or a space. It should be a unique identifier from your LIMS, or modification of the isolate ID.	Illumina HiSeq 2500 [GENEPIO:0100117] XYZ_123345			1.0.0	1.0.0	1.0.0
Sequence information	library preparation kit	GENEPIO:0001450	The name of the DNA library preparation kit used to generate the library being sequenced.	Provide the name of the library preparation kit used.	Nextera XT			1.0.0	1.0.0	1.0.0
Sequence information	genomic target enrichment method	GENEPIO:0100966	The molecular technique used to selectively capture and amplify specific regions of interest from a genome.	Provide the name of the enrichment method by selecting a value from the picklist.	Hybridization capture [GENEPIO:0001950]			1.0.0	1.0.0	1.0.0
Sequence information	genomic target enrichment method details	GENEPIO:0100967	Details that provide additional context to the molecular technique used to selectively capture and amplify specific regions of interest from a genome.	Provide details that are applicable to the method you used.	Enrichment was performed using Illumina Target Enrichment methodology using the Illumina DNA Prep with Enrichment kit.			1.0.0	1.0.0	1.0.0
Sequence information	amplicon per primer scheme name	GENEPIO:0001456	The specification name of the primers (primer sequences, binding positions, fragment size generated etc) used to generate the amplicons to be sequenced.	Provide the name and version of the primer scheme used to generate the amplicons for sequencing.	artc v1			1.0.0	1.0.0	1.0.0
Sequence information	amplicon per primer scheme url	GENEPIO:0102068	A URL linking to detailed information or documentation about the amplicon PCR primer scheme, including specifications such as primer sequences, binding positions, and fragment sizes. This URL should direct users to a source that provides the necessary information to understand the primer scheme used in the dataset.	Provide the url of the primer scheme used to generate the amplicons for sequencing.	https://github.com/infocenterwork/primers-cg-barnes/tree/master/CGV2019V1			1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	amplicon size	GENEPIO:0001449	The length of the amplicon generated by PCR amplification.	Provide the amplicon size expressed in base pairs.	300			1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	raw sequence data processing method	GENEPIO:0001458	The method used for raw data processing (trimming, filtering, adapter trimming, filtering etc).	Provide the software name followed by the version or a link to the software. e.g. Trimmomatic v. 0.36, Porechop v. 0.2.3	Porechop 0.2.3			1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	debiasing method	GENEPIO:0001459	The method used to remove host reads from the pathogen sequence.	Provide the name and version number of the software used to remove host reads.	Nanostripper			1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	genome sequence file name	GENEPIO:0101715	The name of the consensus sequence file.	Provide the name and version number, with the file extension. e.g. assembly.fasta				1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	sequence assembly software name	GENEPIO:0100825	The name of the software used to assemble a sequence.	Provide the name of the software used to assemble the consensus sequence FASTA file or a genome assembly file.	SPAdes Genome Assembler, Canu, wtdbg2, raiet			1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	sequence assembly software version	GENEPIO:0100826	The version of the software used to assemble a sequence.	Provide the version of the software used to assemble the sequence.	3, 15.5			1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	consensus sequence software name	GENEPIO:0001463	The name of the software used to generate the consensus sequence.	Provide the name of the software used to generate the consensus sequence.	ivar			1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	consensus sequence software version	GENEPIO:0001469	The version of the software used to generate the consensus sequence.	Provide the version of the software used to generate the consensus sequence.	1.3			1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	breadth of coverage value	GENEPIO:0001472	The percentage of the reference genome covered by the sequenced data, to a prescribed depth.	Provide value as a percent.	95			1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	depth of coverage value	GENEPIO:0001474	The average number of reads representing a given nucleotide in the reconstructed sequence.	Provide value as a fold of coverage.	400			1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	genome completeness	GENEPIO:0100844	The percentage of expected genes identified in the genome being sequenced. Missing genes are indicated by asterisks in the data.	Provide the genome completeness as a percent (no need to include units).	85			1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	reference genome accession	GENEPIO:0001485	A persistent, unique identifier of a genome database entry.	Provide the accession number of the reference genome.	NC_045512.2			1.0.0	1.0.0	1.0.0

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	Colour Code Legend								
	field name in purple = recommended								
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Bioinformatics and QC metrics	bioinformatics protocol url	GENEPIO.0001489	A description of the overall bioinformatics strategy used.	Further details regarding the methods used to process raw data, and/or generate assemblies, and/or generate consensus sequences can. This information can be provided in a separate document. Provide the name and version number of the protocol, or a GitHub link to a pipeline or workflow.	https://github.com/biasecml/bcov2015-udf-csl		1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	quality control method name	GENEPIO.0100557	The name of the method used to assess whether a sequence passed a predetermined quality control threshold.	Providing the name of the method used for quality control is very important for interpreting the rest of the QC information. Method names can be provided as the name of a pipeline or a link to a GitHub repository. Multiple methods should be listed and separated by a semi-colon. Do not include QC tags in other fields if no method name is provided.	ncov-Hoobs		1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	quality control method version	GENEPIO.0100558	The version number of the method used to assess whether a sequence passed a predetermined quality control threshold.	Methods updates can make big differences to their outputs. The version can be expressed using whatever convention the developer implements (e.g. date, semantic versioning). Multiple methods can be used and their version numbers in the order as the methods names. Separate the version numbers using a semi-colon.	1.2.3		1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	quality control determination	GENEPIO.0100559	The determination of a quality control assessment.	Select a value from the pick list provided. If a desired value is missing, submit a new term request to the PHAAGE QC Tag GitHub issue tracker using the New Term Request form.	Sequence passed quality control [GENEPIO.0100563]		1.0.0	1.0.0	1.0.0
Bioinformatics and QC metrics	quality control issues	GENEPIO.0100560	The reason contributing to, or causing, a low quality determination in a quality control assessment.	Tag a value from the pick list provided. If a desired value is missing, submit a new term request to the PHAAGE QC Tag GitHub issue tracker using the New Term Request form.	Low average genome coverage [GENEPIO.0100570]		1.0.0	1.0.0	1.0.0
Taxonomic Identification information	quality control details	GENEPIO.0100561	The details surrounding a low quality determination in a quality control assessment.	Provide notes or details regarding QC results using free text.	CT value of 39, Low viral load, Low DNA concentration after amplification.		1.0.0	1.0.0	1.0.0
Taxonomic Identification information	taxonomic identification process	GENEPIO.0100563	The type of planned process by which an organismal entity is associated with a taxon or taxa.	Provide the type of method used to determine the taxonomic identity of the organism by selecting a value from the pick list. If the information is unknown or cannot be provided, leave blank or provide a null value.	PCR assay [OBI0002740]		1.0.0	1.0.0	1.0.0
Taxonomic Identification information	read mapping software name	GENEPIO.0100832	The name of the software used to map sequence reads to a reference genome or set of reference genes.	Provide the name of the read mapping software.	Bovli2, BWA-MEM, TopHat		1.0.0	1.0.0	1.0.0
Taxonomic Identification information	read mapping software version	GENEPIO.0100833	The version of the software used to map sequence reads to a reference genome or set of reference genes.	Provide the version number of the read mapping software.	2.5.1		1.0.0	1.0.0	1.0.0
Taxonomic Identification information	taxonomic reference database name	GENEPIO.0100834	The name of the taxonomic reference database used to identify the organism.	Provide the name of the taxonomic reference database.	NCBITaxon		1.0.0	1.0.0	1.0.0
Taxonomic Identification information	taxonomic reference database version	GENEPIO.0100835	The version of the taxonomic reference database used to identify the organism.	Provide the version number of the taxonomic reference database.	1.3		1.0.0	1.0.0	1.0.0
Taxonomic Identification information	viral type	GENEPIO.0102174	Type designation of the virus at the species/serotype/genogroup level (scheme-dependent).	Provide the type designation. Use when the scheme defines a 'type'.	Norovirus GII		1.0.0	1.0.0	1.0.0
Taxonomic Identification information	viral subtype	GENEPIO.0102175	The designation of the subtype of a virus (scheme-dependent).	Provide the subtype designation.	H5N1		1.0.0	1.0.0	1.0.0
Taxonomic Identification information	viral lineage	GENEPIO.0102176	The phylogenetic lineage label assigned to the virus.	Provide the lineage using the appropriate classification scheme.	BA.2.86		1.0.0	1.0.0	1.0.0
Taxonomic Identification information	viral clade	GENEPIO.0102177	The phylogenetic clade label assigned to the virus.	Provide the clade using the appropriate classification scheme. WHO/IEA/FAO H5 clade nomenclature, should be included here. Multiple entries should be captured if more than one system applies.	H5 clade: 2.3.4.4b		1.0.0	1.0.0	1.0.0
Taxonomic Identification information	viral classification scheme	GENEPIO.0102178	The name of the scheme used to define the viral type/subtype/lineage	Include all schemes used to classify the type.	Pango		1.0.0	1.0.0	1.0.0
Taxonomic Identification information	lineage/clade analysis software name	GENEPIO.0001501	The name of the software used to determine the lineage/clade.	Provide the name of the software used to determine the lineage/clade.	Freyja		1.0.0	1.0.0	1.0.0
Target and diagnostic measurements	lineage/clade analysis software version	GENEPIO.0001502	The version of the software used to determine the lineage/clade.	Provide the version of the software used to determine the lineage/clade.	1.5.0		1.0.0	1.0.0	1.0.0

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Target and diagnostic measurements	Colour Code Legend field name in yellow = required field name in purple = recommended field name in white = optional									IMPORTANT: Only labels and/or IDs will be deprecated, always with replacement version provided. If a term changes in its meaning, a new term will be created.
	target taxonomic name 1	GENEPIO:0102049		The target taxonomic name refers to the specific species, genus, or other taxonomic classification that the assay or diagnostic test is designed to identify, detect, or characterize. This may include the full binomial name (Genus species) or a higher taxonomic rank (e.g., genus, family) when species-level identification is not applicable. Avoid abbreviations. Search for taxonomic names here: ncbi.nlm.nih.gov/taxonomy.	Respiratory syncytial virus		1.0.0	1.0.0	1.0.0	
	assay target name 1	GENEPIO:0102052	The name of the assay target used in the diagnostic RT-PCR test.	The specific genomic region, sequence, or variant targeted by the assay in a diagnostic RT-PCR test. This may include parts of a gene, non-coding regions, or other genetic elements that serve as a marker for detecting the presence of a pathogen or other relevant entities.	G2R (Li et al., 2010) assays (includes G2R_WA, C3L, and G2R_G).		1.0.0	1.0.0	1.0.0	
	gene symbol 1	GENEPIO:0102041	The symbol of the gene used in the diagnostic RT-PCR test.	Provide the full name of the gene used in the test. Standardized gene names can be found in the Gene Ontology using this look-up service: https://bit.ly/2Sx1tLj Select a value from the pick list provided, to describe whether a target was determined to be present or absent within a sample.	gryA		1.0.0	1.0.0	1.0.0	
	diagnostic target presence 1	GENEPIO:0100962	The binary value of the result from a diagnostic test.	Provide the numerical result of a diagnostic test (no need to include units). This includes per Ct values.	diagnostic target present		1.0.0	1.0.0	1.0.0	
	diagnostic measurement value 1	GENEPIO:0100963	The value of the result from a diagnostic test.	Select a value from the pick list provided, to describe the units of the given diagnostic test.	PCR quantification cycle (Ct value)		1.0.0	1.0.0	1.0.0	
	diagnostic measurement unit 1	GENEPIO:0100964	The unit of the result from a diagnostic test.	Select a value from the pick list provided to describe the method used for a given diagnostic test. Use this to capture the type of PCR method used in quantifying the pathogen.	qPCR		1.0.0	1.0.0	1.0.0	
	diagnostic measurement method 1	GENEPIO:0100965	The method by which a diagnostic result was determined.	The target taxonomic name refers to the specific species, genus, or other taxonomic classification that the assay or diagnostic test is designed to identify, detect, or characterize. This may include the full binomial name (Genus species) or a higher taxonomic rank (e.g., genus, family) when species-level identification is not applicable. Avoid abbreviations. Search for taxonomic names here: ncbi.nlm.nih.gov/taxonomy .	Respiratory syncytial virus		1.0.0	1.0.0	1.0.0	
	target taxonomic name 2	GENEPIO:0102050					1.0.0	1.0.0	1.0.0	
Target and diagnostic measurements	assay target name 2	GENEPIO:0102038	The name of the assay target used in the diagnostic RT-PCR test.	The specific genomic region, sequence, or variant targeted by the assay in a diagnostic RT-PCR test. This may include parts of a gene, non-coding regions, or other genetic elements that serve as a marker for detecting the presence of a pathogen or other relevant entities.	G2R (Li et al., 2010) assays (includes G2R_WA, C3L, and G2R_G).		1.0.0	1.0.0	1.0.0	
Target and diagnostic measurements	gene symbol 2	GENEPIO:0102042	The symbol of the gene used in the diagnostic RT-PCR test.	Provide the full name of the gene used in the test. Standardized gene names can be found in the Gene Ontology using this look-up service: https://bit.ly/2Sx1tLj Select a value from the pick list provided, to describe whether a target was determined to be present or absent within a sample.	gyrase A		1.0.0	1.0.0	1.0.0	
Target and diagnostic measurements	diagnostic target presence 2	GENEPIO:0100962	The binary value of the result from a diagnostic test.	Provide the numerical result of a diagnostic test (no need to include units).	diagnostic target present		1.0.0	1.0.0	1.0.0	
Target and diagnostic measurements	diagnostic measurement value 2	GENEPIO:0100963	The value of the result from a diagnostic test.	Select a value from the pick list provided, to describe the units of the given diagnostic test.	cycle threshold (Ct)		1.0.0	1.0.0	1.0.0	
Target and diagnostic measurements	diagnostic measurement unit 2	GENEPIO:0100964	The unit of the result from a diagnostic test.	Select a value from the pick list provided to describe the method used for a given diagnostic test. Use this to capture the type of PCR method used in quantifying the pathogen.	qPCR		1.0.0	1.0.0	1.0.0	
Target and diagnostic measurements	diagnostic measurement method 2	GENEPIO:0100965	The method by which a diagnostic result was determined.	The target taxonomic name refers to the specific species, genus, or other taxonomic classification that the assay or diagnostic test is designed to identify, detect, or characterize. This may include the full binomial name (Genus species) or a higher taxonomic rank (e.g., genus, family) when species-level identification is not applicable. Avoid abbreviations. Search for taxonomic names here: ncbi.nlm.nih.gov/taxonomy .	Respiratory syncytial virus		1.0.0	1.0.0	1.0.0	
Target and diagnostic measurements	target taxonomic name 3	GENEPIO:0102051					1.0.0	1.0.0	1.0.0	
Target and diagnostic measurements	assay target name 3	GENEPIO:0102039	The name of the assay target used in the diagnostic RT-PCR test.	The specific genomic region, sequence, or variant targeted by the assay in a diagnostic RT-PCR test. This may include parts of a gene, non-coding regions, or other genetic elements that serve as a marker for detecting the presence of a pathogen or other relevant entities.	G2R (Li et al., 2010) assays (includes G2R_WA, C3L, and G2R_G).		1.0.0	1.0.0	1.0.0	

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Target and diagnostic measurements	gene symbol 3	GENEPIO:0102043	The symbol of the gene used in the diagnostic RT-PCR test.	Provide the full name of the gene used in the test. Standardized gene names can be found in the Gene Ontology using this look-up service: https://bit.ly/2Sqt1Ld	gyrase A			1.0.0	1.0.0	
Target and diagnostic measurements	diagnostic target presence 3	GENEPIO:0100962	The binary value of the result from a diagnostic test.	Select a value from the pick list provided, to describe whether a target was determined to be present or absent within a sample.	diagnostic target present			1.0.0	1.0.0	
Target and diagnostic measurements	diagnostic measurement value 3	GENEPIO:0100963	The value of the result from a diagnostic test.	Provide the numerical result of a diagnostic test (no need to include units).				1.0.0	1.0.0	
Target and diagnostic measurements	diagnostic measurement unit 3	GENEPIO:0100964	The unit of the result from a diagnostic test.	Select a value from the pick list provided, to describe the units of the given diagnostic test.	cycle threshold (Ct)			1.0.0	1.0.0	
Contributor acknowledgement	diagnostic measurement method 3	GENEPIO:0100965	The method by which a diagnostic result was determined.	Select a value from the pick list provided to describe the method used for a given diagnostic test. Use this to capture the type of PCR method used in quantifying the pathogen.	qPCR			1.0.0	1.0.0	
Contributor acknowledgement	authors	GENEPIO:0001517	Names of individuals contributing to the processes of sample collection, sequence generation, analysis, and data submission.	Include the first and last names of all individuals that should be attributed, separated by a semicolon.	tejinder Singh; Fei Hu; Joe Blugs			1.0.0	1.0.0	
Contributor acknowledgement	citation URL	GENEPIO:0102208						1.0.0	1.0.0	

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